

Information Provenance and Conservative Enrichment in Prospective Health

A Machine-Checked Finite Forest Benchmark

Zed James

Paper IV in the series: *Health, Formally Defined*.

© 2026 Zed James. This work is licensed under the Creative Commons
Attribution-NonCommercial-NoDerivatives 4.0 International License.

Preprint – October 9, 2026.
doi:10.5281/zenodo.23268803.

Abstract

Prospective Health evaluates present organizational realization together with adequate viable-continuation capacity under declared conditions. Papers I–III establish the formal definition, contextual semantics, and query-relative representation theory. This paper supplies a machine-checked finite benchmark whose information architecture can be inspected exhaustively.

A five-state forest system is represented through two equivalent history encodings. The original encoding places contextual admissibility in dependent history constructors; the second admits permissive candidate histories and applies a nontrivial lawfulness predicate. We prove equivalence of lawful histories, preservation of viable continuations, and invariance of collected capacity and Health. The permissive encoding exposes the exact candidate–lawful–viable response pipeline.

Complete detailed-capacity, original-capacity, and declared-Health profiles induce the equality-kernel ladder

$$\boxed{B \mid P \mid R \mid A \mid F \longrightarrow B \mid P \mid \{R, A\} \mid F \longrightarrow \{B, R, A\} \mid P \mid F.}$$

Every lost distinction is classified according to whether response representation erases it or the declared Health language leaves it unused. A domain-neutral conservative-refinement lemma establishes preservation of all established Health judgments when richer capacity projects exactly to established capacity and adequacy is pulled back through that projection. A strict forest instance recovers a previously erased distinction while maintaining all established judgments.

These are exact guarantees relative to the declared finite histories, dynamics, constitution, observers, and query language. Capacity in this benchmark is the set of attained response atoms. Equal attained supports alone do not determine the probability-weighted capacity or failure mass of an independently specified stochastic law. This benchmark therefore provides a formal reference for the companion empirical-synthetic forest measurement study, which tests reconstruction, model discrepancy, representation adequacy, and numerical decision resolution under incomplete observations.

1 Introduction

For measurement-oriented scientific use, the information requirements of a formal health definition must be traceable through an actual domain specification. In particular, one should be able to determine which prospective responses were considered, which were ruled out by domain law, which preserved the organization, which distinctions survived observation, which were removed by representation, and which retained distinctions were unused by the declared health questions.

Papers I–III establish the general architecture. Paper I defines prospective health from present realization and adequate continuation capacity [1]. Paper II develops contextual licensing and relation-generated transport [2]. Paper III places health-relevant representations in an equality-kernel refinement order and proves that a representation is sufficient for a declared query family exactly when it refines that family’s canonical quotient [3]. For a full state-level health family H , Paper III proves

$$\text{HealthObservationSufficient}(H, M) \iff M \succeq q_H. \quad (1)$$

Equation (1) is inherited here and supplies the generic observation criterion used throughout the benchmark.

The question addressed here is different:

Within one coherent domain benchmark, where are distinctions
admitted, filtered, compressed, ignored, and recoverable?

A finite forest system is useful because its complete semantic surface can be audited. The benchmark contains baseline, stressed, recovered, compositionally adapted, and collapsed states; mild and severe scenarios; short and long horizons; disturbance, regeneration, adaptation, collapse, and canopy-turnover event forms; and two declared health requirements. The finite size permits complete censuses and exact equality-kernel classifications and contains several independent information-loss mechanisms.

The central result is an exact provenance ladder:

$$B \mid P \mid R \mid A \mid F \longrightarrow B \mid P \mid \{R, A\} \mid F \longrightarrow \{B, R, A\} \mid P \mid F. \quad (2)$$

The first arrow is representational compression: recovery and compositional adaptation have distinct detailed response provenance and equal original two-atom capacity. The second arrow is query-relative compression: baseline retains a different original capacity profile from recovered and adapted. Every declared Health query agrees across those states.

This yields two different diagnoses of equal health answers. A distinction may still be present in the representation and remain unused by the current query language. A distinction may also have been erased earlier by the representation itself. The remedies differ. Query enrichment can use retained information. Representation enrichment is required after erasure.

The paper’s constructive response to the second case is a conservative-enrichment principle. A richer observer may project exactly to an established observer, and established adequacy may be pulled back through that projection. The preservation lemma makes this compatibility condition explicit. The strict forest instance restores the recovered/adapted distinction, preserves arbitrary established adequacy, supports an additional adaptation-sensitive query, and preserves every earlier Health judgment.

1.1 Contributions and inherited results

To keep the series boundary explicit, Table 1 separates inherited theory from new results.

Result	Status in this paper
M sufficient for H iff $M \succeq q_H$	Inherited from Paper III; instantiated here.
Canonical query-visible capacity and state quotients	Inherited construction; concrete forest kernels computed here.
Equivalent indexed-history and permissive-candidate encodings	New forest benchmark result.
Candidate $\rightarrow$ lawful $\rightarrow$ viable response census	New forest benchmark result.
Exact $5 \rightarrow 4 \rightarrow 3$ information ladder	New forest benchmark result.
Complete localization of every pair lost at the two information steps	New forest benchmark result.
Generic conservative-capacity-refinement lemma	New domain-neutral preservation lemma extracted for reuse.
Strict forest enrichment preserving arbitrary established adequacy	New strict benchmark result; this carries the constructive weight of the enrichment argument.

Table 1: Series ownership of the principal results.

The forest benchmark also retains supporting instances involving context, support, transport, cost, control, route structure, probability, robustness, and component turnover. These appear later as integration checks and in the formal appendix. The semantic pipeline, exact kernels, and conservative-enrichment results establish the provenance argument.

The finite benchmark has independent value after an empirical-synthetic implementation has been constructed. It proves properties of fully declared semantics that successful computation alone cannot imply: equivalence between two placements of admissibility, exhaustive localization of distinctions lost to observation and query evaluation, and backward compatibility under an exact capacity projection. Finite enumeration is the method that makes those claims complete rather than merely illustrative. The five state labels form a controlled formal test domain, not an ecological state taxonomy.

The companion empirical-synthetic forest study (Paper V) instantiates a different problem: incomplete census measurements require state reconstruction; provisional demographic laws generate futures; and stochastic, numerical, and semantic uncertainties affect the resulting Health decisions. The formal results here identify precise compatibility and sufficiency obligations, while the companion study tests the reliability of particular empirical approximations. Both manuscripts specify their respective objects independently.

2 Local formal vocabulary

This section restates the definitions needed to read the paper independently of Papers I–III.

2.1 Prospective health

Fix state, scenario, and horizon carriers X, D, T . A dynamics supplies a dependent history family

$$\text{Hist}(d, t, x, y)$$

and a lawfulness predicate on each history. A constitutive specification supplies

$$\text{Realizes} : X \rightarrow \text{Prop}$$

and

$$\text{Continues} : \text{Hist}(d, t, x, y) \rightarrow \text{Prop}.$$

For source state x , scenario d , and horizon t , the viable continuation carrier contains exactly histories equipped with proofs of both lawfulness and constitutive continuation:

$$\mathcal{V}_{d,t}(x) = \{(y, \gamma) \mid \gamma \in \text{Hist}(d, t, x, y), \text{Lawful}(\gamma), \text{Continues}(\gamma)\}. \quad (3)$$

A capacity observer assigns an atom to each viable continuation and collects all attained atoms into a capacity representation:

$$\text{Capacity}(x; d, t) = \text{collect}\{o(v) : v \in \mathcal{V}_{d,t}(x)\}. \quad (4)$$

For requirement r , adequacy sees capacity alone. Health is

$$\boxed{\text{Health}_r(x; d, t) \iff \text{Realizes}(x) \wedge \text{Adequate}_r(\text{Capacity}(x; d, t))}. \quad (5)$$

2.2 Attained-response support and weighted capacity

The capacity collector in Equation (4) is *set-valued*: it returns the set of response atoms attained by viable histories. It carries no distribution on the histories or their observed response atoms. If a separate stochastic history law μ is imposed, then its original-law viable mass is a different object,

$$m_{V,\mu}(x; d, t) = \mu_{x,d,t}(\mathcal{V}_{d,t}(x)), \quad m_{F,\mu}(x; d, t) = 1 - m_{V,\mu}(x; d, t). \quad (6)$$

The displayed complement aggregates all histories outside viability; finer accounting may separate inadmissible histories from lawful noncontinuation. For example, two laws can both attain only the viable atom $\{S\}$, while assigning viable mass 0.9 and 0.2, and failure-complement mass 0.1 and 0.8, respectively. Thus

$$\text{Capacity}_1^{\text{set}} = \text{Capacity}_2^{\text{set}} = \{S\} \not\Rightarrow m_{V,\mu_1} = m_{V,\mu_2}.$$

The encoding-equivalence and conservative-refinement theorems in this paper concern the declared set-valued capacities. Transferring such preservation to probability-sensitive capacity requires compatible path measures and preservation of successful and failed-history mass, beyond equality of attainable response supports.

2.3 Representation refinement and sufficiency

For encodings

$$e_1 : X \rightarrow O_1, \quad e_0 : X \rightarrow O_0,$$

define equality-kernel refinement by

$$\boxed{e_1 \succeq e_0 \iff \forall x, y, e_1(x) = e_1(y) \implies e_0(x) = e_0(y)}. \quad (7)$$

Thus e_1 retains every distinction retained by e_0 , possibly together with additional distinctions.

For a query family

$$A : Q \times X \rightarrow \mathbf{Prop},$$

a representation $e : X \rightarrow O$ is sufficient when equal representation values imply agreement on every query:

$$\text{RepresentationSufficient}(A, e) \iff \forall x, y, e(x) = e(y) \implies \forall q, A(q, x) \leftrightarrow A(q, y). \quad (8)$$

The induced query equivalence is

$$x \sim_A y \iff \forall q, A(q, x) \leftrightarrow A(q, y),$$

with quotient map

$$q_A : X \rightarrow X / \sim_A.$$

Paper III proves

$$\text{RepresentationSufficient}(A, e) \iff e \succeq q_A.$$

For a single predicate $P : X \rightarrow \mathbf{Prop}$, write $\text{PredicateSufficient}(e, P)$ for the analogous one-query condition.

2.4 Localization of information loss

For a fine representation e_1 and a coarser representation e_0 , define

$$\boxed{\text{CollapsedBetween}(e_1, e_0; x, y) \iff e_1(x) \neq e_1(y) \wedge e_0(x) = e_0(y)}. \quad (9)$$

This predicate identifies the stage at which a particular distinction disappears. ¹

2.5 Exact relation-level determinacy

For a relation $R : X \rightsquigarrow Y$, source representation $S : X \rightarrow O_S$, and target representation $T : Y \rightarrow O_T$, define

$$\text{Determinate}(R, S, T) \iff \forall x, x', y, y', R(x, y) \wedge R(x', y') \wedge S(x) = S(x') \implies T(y) = T(y'). \quad (10)$$

This definition appears in the later integration section; the provenance results below are established independently of transport assumptions.

¹Formal counterpart: `lean/Tier33/Health/Representation/ConservativeRefinement.lean::CollapsedBetween;collapsedBetween_symm.`

3 The finite forest benchmark

3.1 States, contexts, and original indexed histories

The state carrier is

$$X = \{B, P, R, A, F\},$$

where

$$B = \text{baseline}, \quad P = \text{stressed}, \quad R = \text{recovered}, \quad A = \text{adapted}, \quad F = \text{collapsed}.$$

The scenario carrier is

$$D = \{\text{mild}, \text{severe}\},$$

and the horizon carrier is

$$T = \{\text{now}, \text{long}\}.$$

The original dependent history datatype contains six event forms:

Event	Indexed availability	Endpoint
stay	every context and state	$x \rightarrow x$
disturb	mild, either horizon	$B \rightarrow P$
regenerate	mild/long	$P \rightarrow R$
adaptComposition	mild/long	$P \rightarrow A$
severeCollapse	severe/long	$P \rightarrow F$
canopyTurnover	mild/long	$B \rightarrow B$

The datatype itself therefore carries contextual admissibility information. ²

The realization criterion is

$$\boxed{\text{Realizes}(x) \iff x \neq F.} \tag{11}$$

Continuation is endpoint constitutive:

$$\boxed{\text{Continues}(\gamma) \iff \text{Realizes}(\text{target}(\gamma)).} \tag{12}$$

Hence severe collapse is lawful in its indexed context and fails continuation. ³

3.2 Original capacity and declared requirements

The original capacity atom carrier is

$$\mathcal{A} = \{E, S\},$$

with E = exposed and S = stable. The observer reads histories and the collector forms the set of attained atoms.

The complete capacity census, derived from the full indexed history family, is:

²Formal counterpart: `lean/Tier33/Health/Forest/Core.lean::History`.

³Formal counterpart:

`lean/Tier33/Health/Forest/Core.lean::severeCollapse_lawful;severeCollapse_not_continuing`.

State	mild/now	mild/long	severe/now	severe/long
B	$\{S, E\}$	$\{S, E\}$	$\{S\}$	$\{S\}$
P	$\{E\}$	$\{E, S\}$	$\{E\}$	$\{E\}$
R	$\{S\}$	$\{S\}$	$\{S\}$	$\{S\}$
A	$\{S\}$	$\{S\}$	$\{S\}$	$\{S\}$
F	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

4

The declared requirements are

$$r_{\text{any}}, \quad r_{\text{stable}},$$

with

$$\text{Adequate}_{r_{\text{any}}}(c) \iff c \neq \emptyset, \quad \text{Adequate}_{r_{\text{stable}}}(c) \iff S \in c.$$

These generate requirement-, scenario-, and horizon-relative Health on the stressed state. ⁵

The finite benchmark is complete relative to its declared datatypes and query family. Ecological completeness and prospective validation require independently licensed empirical models and evidence.

4 Encoding invariance: two ways to place admissibility

The original history datatype encodes contextual admissibility through constructor availability. This design makes the forest’s original lawfulness predicate extensionally true on every inhabitant of the indexed datatype. A second encoding exposes the same constraints as an explicit filter.

4.1 Permissive candidate histories

Define a candidate history family with the same six event forms available across every scenario and horizon, with their source and target states retained:

$$\text{CandidateHist}(d, t, x, y).$$

The contextual law is

$$\begin{aligned} \text{Lawful}(\text{disturb}) &\iff d = \text{mild}, \\ \text{Lawful}(\text{regenerate}) &\iff d = \text{mild} \wedge t = \text{long}, \\ \text{Lawful}(\text{adaptComposition}) &\iff d = \text{mild} \wedge t = \text{long}, \\ \text{Lawful}(\text{severeCollapse}) &\iff d = \text{severe} \wedge t = \text{long}, \\ \text{Lawful}(\text{canopyTurnover}) &\iff d = \text{mild} \wedge t = \text{long}. \end{aligned}$$

⁴Formal counterpart: `lean/Tier33/Health/Forest/Core.lean::capacity_eq_expected`.

⁵Formal counterpart: `lean/Tier33/Health/Forest/HealthLanguage.lean::forest_health_normal_form; requirement_changes_forest_health; scenario_changes_forest_health; horizon_changes_forest_health`.

Stay remains lawful at its typed equal endpoint. ⁶

The lawfulness filter is now nontrivial. For example,

$$\neg \text{Lawful}(\text{regenerate}; \text{mild}, \text{now}), \quad \text{Lawful}(\text{regenerate}; \text{mild}, \text{long}), \quad (13)$$

and

$$\neg \text{Lawful}(\text{collapse}; \text{mild}, \text{long}), \quad \text{Lawful}(\text{collapse}; \text{severe}, \text{long}). \quad (14)$$

⁷

The candidate constitution uses the same realization criterion as Equation (11). Lawful severe/long collapse therefore fails continuation:

$$\boxed{\text{Lawful}(\text{collapse}; \text{severe}, \text{long}) \wedge \neg \text{Continues}(\text{collapse}; \text{severe}, \text{long})}. \quad (15)$$

⁸

4.2 Exact equivalence of lawful encodings

The two history encodings represent the same admissible forest semantics.

Theorem 4.1 (Lawful-history encoding equivalence). *For every d, t, x, y ,*

$$\boxed{\{\gamma : \text{CandidateHist}(d, t, x, y) \mid \text{Lawful}(\gamma)\} \simeq \text{Hist}(d, t, x, y)}. \quad (16)$$

⁹

The equivalence preserves endpoints and lifts to viable continuations:

$$\boxed{\mathcal{V}_{d,t}^{\text{cand}}(x) \simeq \mathcal{V}_{d,t}^{\text{orig}}(x)}. \quad (17)$$

¹⁰

The observer commutes across the equivalence. Consequently,

$$\boxed{\text{Capacity}_{\text{cand}}(x; d, t) = \text{Capacity}_{\text{orig}}(x; d, t)} \quad (18)$$

for every state and context, and

$$\boxed{\text{CandidateForestHealth}_r(x; d, t) \iff \text{ForestHealth}_r(x; d, t)}. \quad (19)$$

⁶Formal counterpart:

`lean/Tier33/Health/Forest03/CandidateSemantics.lean::CandidateHistory;CandidateLawful.`

⁷Formal counterpart:

`lean/Tier33/Health/Forest03/CandidateSemantics.lean::candidate_regenerate_now_not_lawful;candidate_regenerate_long_lawful;candidate_collapse_mild_not_lawful;candidate_collapse_severe_lawful.`

⁸Formal counterpart: `lean/Tier33/Health/Forest03/CandidateSemantics.lean::`

`candidate_collapse_severe_lawful;candidate_collapse_not_continuing.`

⁹Formal counterpart: `lean/Tier33/Health/Forest03/LawfulEncodingEquivalence.lean::`

`lawfulCandidateEquivHistory;candidate_history_roundtrip;original_history_roundtrip.`

¹⁰Formal counterpart: `lean/Tier33/Health/Forest03/LawfulEncodingEquivalence.lean::`
`candidateViableEquivOriginal;candidate_viable_roundtrip;original_viable_roundtrip.`

Equations (16)–(19) establish encoding invariance for this benchmark. Contextual admissibility may be carried by dependent constructor availability or by an explicit lawfulness predicate, with the downstream Health theory preserved whenever the lawful history carriers are equivalent and observation is preserved.

5 Semantic provenance: candidate, lawful, and viable responses

Encoding invariance allows the permissive model to expose the filtering stages directly.

5.1 Response-kind carriers

Let

$$\mathcal{K} = \{\text{stay, disturb, regenerate, adapt, collapse, turnover}\}.$$

For each state and context define the response-kind sets

$$K_{\text{cand}}(x; d, t), \quad K_{\text{law}}(x; d, t), \quad K_{\text{viable}}(x; d, t).$$

They collect, respectively, kinds represented by candidate histories, lawful candidate histories, and viable candidate continuations.

The formal development derives complete expected-set theorems for all three stages, uniformly over every state, scenario, and horizon. ¹²

For all x, d, t ,

$$\boxed{K_{\text{viable}}(x; d, t) \subseteq K_{\text{law}}(x; d, t) \subseteq K_{\text{cand}}(x; d, t).} \quad (20)$$

¹³

Both filters are strict in the benchmark. At stressed/mild/now,

$$\boxed{\text{regenerate} \in K_{\text{cand}} \setminus K_{\text{law}}.} \quad (21)$$

At stressed/severe/long,

$$\boxed{\text{collapse} \in K_{\text{law}} \setminus K_{\text{viable}}.} \quad (22)$$

¹⁴

The first loss is contextual legality. The second is constitutive viability. The two predicates therefore perform distinct semantic work in one and the same forest model.

¹¹Formal counterpart: `lean/Tier33/Health/Forest03/LawfulEncodingEquivalence.lean::candidate_observation_commutates; candidate_capacity_eq_original; candidateHealth_iff_originalHealth.`

¹²Formal counterpart: `lean/Tier33/Health/Forest03/ResponsePipeline.lean::candidateResponseKinds_eq_expected; lawfulResponseKinds_eq_expected; viableResponseKinds_eq_expected; responseKinds_complete_census.`

¹³Formal counterpart: `lean/Tier33/Health/Forest03/ResponsePipeline.lean::responseKinds_inclusions.`

¹⁴Formal counterpart: `lean/Tier33/Health/Forest03/ResponsePipeline.lean::regeneration_removed_by_lawfulness; collapse_removed_by_constitution.`

5.2 From viable responses to observed capacity

The detailed observer introduced below retains source-sensitive stable provenance. A response-kind map sends viable kinds exactly to the detailed capacity:

$$\kappa_x[K_{\text{viable}}(x; d, t)] = C_D(x; d, t), \quad (23)$$

where κ_x retains source state for the stay kind. ¹⁵

Composing κ_x with the coarse forgetting map π gives the original capacity exactly:

$$(\pi \circ \kappa_x)[K_{\text{viable}}(x; d, t)] = C_C(x; d, t). \quad (24)$$

¹⁶

Equations (20)–(24) give a continuous semantic provenance chain from candidate event forms to the capacity representation used by Health.

6 Detailed and coarse capacity

6.1 Detailed response representation

The original atom S intentionally merges three stable response modes. Define a richer atom carrier

$$\mathcal{A}_D = \{E, S_B, S_R, S_A\},$$

where

$$S_B = \text{baselineStable}, \quad S_R = \text{recoveredStable}, \quad S_A = \text{adaptedStable}.$$

The detailed observer uses the same dynamics, constitution, and entire viable-continuation family. Its global census is:

State	mild/now	mild/long	severe/now	severe/long
B	$\{S_B, E\}$	$\{S_B, E\}$	$\{S_B\}$	$\{S_B\}$
P	$\{E\}$	$\{E, S_R, S_A\}$	$\{E\}$	$\{E\}$
R	$\{S_R\}$	$\{S_R\}$	$\{S_R\}$	$\{S_R\}$
A	$\{S_A\}$	$\{S_A\}$	$\{S_A\}$	$\{S_A\}$
F	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

¹⁷

¹⁵Formal counterpart: `lean/Tier33/Health/Forest03/ResponsePipeline.lean::kindDetailedAtom;viable_responseKinds_map_detailed_capacity.`

¹⁶Formal counterpart: `lean/Tier33/Health/Forest03/ResponsePipeline.lean::viable_responseKinds_map_original_capacity.`

¹⁷Formal counterpart: `lean/Tier33/Health/Forest02/DetailedCapacity.lean::detailedCapacity_eq_expected.`

Define the forgetting map

$$\pi(E) = E, \quad \pi(S_B) = \pi(S_R) = \pi(S_A) = S,$$

extended to capacity sets by direct image. For every state and context,

$$\boxed{\pi(C_D(x; d, t)) = C_C(x; d, t).} \quad (25)$$

18

7 The exact information ladder

Define complete context profiles

$$D(x) = [(d, t) \mapsto C_D(x; d, t)], \quad (26)$$

$$C(x) = [(d, t) \mapsto C_C(x; d, t)], \quad (27)$$

$$H(x) = [q \mapsto \text{ForestHealth}_q(x)]. \quad (28)$$

Here H ranges over both declared requirements, both scenarios, and both horizons.

The profiles satisfy

$$\boxed{D \succeq C \succeq H.} \quad (29)$$

Both reverse refinements fail. ¹⁹

7.1 Detailed profile: five classes

Theorem 7.1 (Detailed-profile kernel). *For all forest states x, y ,*

$$\boxed{D(x) = D(y) \iff x = y.} \quad (30)$$

20

Thus the detailed capacity profile has equality classes

$$\boxed{B \mid P \mid R \mid A \mid F.} \quad (31)$$

7.2 Original capacity profile: four classes

Theorem 7.2 (Coarse-capacity kernel). *For all x, y ,*

$$\boxed{C(x) = C(y) \iff x = y \vee (x = R \wedge y = A) \vee (x = A \wedge y = R).} \quad (32)$$

¹⁸Formal counterpart:

`lean/Tier33/Health/Forest02/ConservativeEnrichment.lean::project_detailed_capacity.`

¹⁹Formal counterpart: `lean/Tier33/Health/Forest03/InformationLadder.lean::information_ladder_refines ; coarseProfile_does_not_refine_detailedProfile ; healthSignature_does_not_refine_coarseProfile.`

²⁰Formal counterpart:

`lean/Tier33/Health/Forest03/InformationLadder.lean::detailedProfile_eq_iff_state_eq.`

21

Therefore the original capacity profile has exactly four classes:

$$\boxed{B \mid P \mid \{R, A\} \mid F.} \quad (33)$$

7.3 Declared Health profile: three classes

The complete Health signature satisfies

$$\boxed{H(x) = H(y) \iff h_H(x) = h_H(y),} \quad (34)$$

where

$$h_H : X \rightarrow \text{FullForestHealthCode}$$

assigns

$$\{B, R, A\} \mapsto \text{allDeclaredHealthy}, \quad P \mapsto \text{contextDependent}, \quad F \mapsto \text{nonRealizing}.$$

22

Hence

$$\boxed{\{B, R, A\} \mid P \mid F.} \quad (35)$$

Equations (31), (33), and (35) give the exact information ladder:

$$\boxed{\begin{array}{c} B \mid P \mid R \mid A \mid F \\ \downarrow \\ B \mid P \mid \{R, A\} \mid F \\ \downarrow \\ \{B, R, A\} \mid P \mid F. \end{array}} \quad (36)$$

The first arrow occurs in the observer projection π . The second occurs when the declared Health language consumes the coarse representation.

8 Exact localization of information loss

The ladder permits every distinct-state collision to be assigned to the stage where it first appears.

²¹ Formal counterpart: `lean/Tier33/Health/Forest03/InformationLadder.lean::coarseProfile_eq_iff`.

²² Formal counterpart:

`lean/Tier33/Health/Forest02/FullHealthCode.lean::fullHealthSignature_eq_iff_code_eq;fullHealthCode`.

8.1 Representation-level erasure

Recovered and adapted have distinct detailed profiles and identical coarse profiles:

$$D(R) \neq D(A), \quad C(R) = C(A).$$

Thus

$$\text{CollapsedBetween}(D, C; R, A).$$

The converse classification is exact.

Theorem 8.1 (Complete detailed-to-coarse loss classification). *For all states x, y ,*

$$\boxed{\text{CollapsedBetween}(D, C; x, y) \iff (x = R \wedge y = A) \vee (x = A \wedge y = R).} \quad (37)$$

²³

This is representation-level erasure: the original capacity has already merged the distinction. Consequently every predicate and function of that capacity agrees on recovered and adapted in every context, including arbitrary adequacy installed over the old observer. ²⁴

8.2 Query-invisible retained information

Baseline remains distinct from recovered and adapted at the coarse-capacity level:

$$C(B) \neq C(R), \quad C(B) \neq C(A),$$

The complete Health signature is shared:

$$H(B) = H(R) = H(A).$$

The full classification is again exact.

Theorem 8.2 (Complete coarse-to-Health loss classification). *For all x, y ,*

$$\boxed{\text{CollapsedBetween}(C, H; x, y) \iff (x = B \wedge (y = R \vee y = A)) \vee (y = B \wedge (x = R \vee x = A)).} \quad (38)$$

²⁵

Here the representation retains the distinguishing information. The declared Health language identifies states that agree on its licensed requirements. For example, at mild/now baseline capacity is $\{S, E\}$ and recovered capacity is $\{S\}$, so the predicate

$$P_E(c) \iff E \in c$$

²³Formal counterpart: `lean/Tier33/Health/Forest03/CollapseLocalization.lean::detailed_to_coarse_collapse_iff;distinct_coarse_collision_iff.`

²⁴Formal counterpart: `lean/Tier33/Health/Forest02/CollapseProvenance.lean::no_capacity_predicate_separates_recovered_adapted;no_capacity_function_separates_recovered_adapted;current_observer_health_cannot_distinguish_recovered_adapted.`

²⁵Formal counterpart: `lean/Tier33/Health/Forest03/CollapseLocalization.lean::coarse_to_health_collapse_iff;query_invisible_retained_distinction_iff.`

separates them using information already present in the existing observer representation. ²⁶

The two mechanisms therefore have different formal diagnoses:

$$\boxed{\begin{array}{ll} R/A : & \text{erased by coarse representation,} \\ B/R \text{ and } B/A : & \text{retained by capacity; unused by declared Health.} \end{array}} \quad (39)$$

Within this finite benchmark, Theorems 8.1 and 8.2 exhaust the distinct-state losses at the two stages.

9 Conservative representational refinement

The preceding diagnosis motivates a constructive question: when a new query requires information absent from an established representation, can a richer representation preserve the established Health judgments and expose the needed distinction?

9.1 A generic preservation principle

Fix one dynamics and constitution, with an established capacity observer O_0 and a richer observer O_1 . A conservative capacity refinement supplies a projection

$$\pi : O_1.\text{Cap} \rightarrow O_0.\text{Cap}$$

satisfying

$$\boxed{\pi(\text{Capacity}_{O_1}(x; d, t)) = \text{Capacity}_{O_0}(x; d, t)} \quad (40)$$

for every state, scenario, and horizon. ²⁷

For any established adequacy A on O_0 , define its pullback

$$A^\pi(c) = A(\pi(c)). \quad (41)$$

Lemma 9.1 (Conservative Health preservation). *For every conservative capacity refinement, every old adequacy A , and every x, d, t ,*

$$\boxed{\text{Health}_{O_1, A^\pi}(x; d, t) \iff \text{Health}_{O_0, A}(x; d, t).} \quad (42)$$

Moreover, for each fixed d, t ,

$$\boxed{\text{Capacity}_{O_1}(\cdot; d, t) \succeq \text{Capacity}_{O_0}(\cdot; d, t).} \quad (43)$$

²⁸

²⁶ Formal counterpart: `lean/Tier33/Health/Forest02/CollapseProvenance.lean::baseline_exposedAvailable;recovered_not_exposedAvailable.`

²⁷ Formal counterpart: `lean/Tier33/Health/Representation/ConservativeRefinement.lean::ConservativeCapacityRefinement.`

²⁸ Formal counterpart: `lean/Tier33/Health/Representation/ConservativeRefinement.lean::ConservativeCapacityRefinement.liftAdequacy;ConservativeCapacityRefinement.health_iff;ConservativeCapacityRefinement.capacity_refines.`

Present realization is unchanged because the observers share a constitution. Equation (40) rewrites rich capacity to old capacity under the projection, and the pulled-back adequacy then evaluates the established predicate on that established capacity.

Remark 9.2 (Exact scope of conservative preservation). The theorem fixes a single state carrier, dynamics, lawful/viable-history semantics, and constitution; it varies the observer and uses the exact projection hypothesis (40) for every x, d, t . It therefore proves the stated implication for arbitrary established adequacy without licensing an arbitrary new requirement. If enrichment also changes the state carrier, transition kernel, continuation criterion, or probability law, additional compatibility conditions are needed. Equality of initial structural summaries, matching path seeds, and equality of attained supports do not individually establish the required prospective-capacity commutation.

9.2 Strict forest instance

The detailed forest observer and original observer satisfy the generic structure with

$$\pi = \text{projectDetailedCapacity}.$$

29

For every old adequacy A ,

$$\boxed{\text{Health}_{D, A^\pi}(x; d, t) \iff \text{Health}_{C, A}(x; d, t).} \quad (44)$$

30

The original forest adequacy language is exactly the pullback language, so

$$\boxed{\text{DetailedForestHealth}_r(x; d, t) \iff \text{ForestHealth}_r(x; d, t)} \quad (45)$$

for every declared requirement, state, scenario, and horizon. ³¹

The refinement is strict: $D(R) \neq D(A)$ and $C(R) = C(A)$. Define the additional query

$$P_A(x) \iff S_A \in C_D(x; \text{mild}, \text{now}).$$

At mild/now,

$$\neg P_A(R), \quad P_A(A).$$

The old capacity representation is insufficient for P_A , and the detailed representation is sufficient:

$$\boxed{\neg \text{PredicateSufficient}(C_C(\cdot; \text{mild}, \text{now}), P_A), \quad \text{PredicateSufficient}(C_D(\cdot; \text{mild}, \text{now}), P_A).} \quad (46)$$

32

²⁹ *Formal counterpart:*

`lean/Tier33/Health/Forest03/ConservativeRefinement.lean::forestConservativeRefinement.`

³⁰ *Formal counterpart:*

`lean/Tier33/Health/Forest03/ConservativeRefinement.lean::forest_arbitrary_adequacy_preserved.`

³¹ *Formal counterpart:* `lean/Tier33/Health/Forest03/ConservativeRefinement.lean::`

`forest_liftAdequacy_eq_detailed; forestHealth_preservation_from_generic.`

³² *Formal counterpart:*

`lean/Tier33/Health/Forest02/EnrichmentBoundary.lean::original_capacity_insufficient_for_adaptationMode; detailed_capacity_sufficient_for_adaptationMode; forest_representation_enrichment_boundary.`

Thus the benchmark realizes a strict conservative enrichment:

$$\boxed{\begin{array}{c} \text{additional representational resolution} + \text{exact backward projection} + \text{adequacy pullback} \\ \implies \\ \text{preservation of established Health} \end{array}} \quad (47)$$

10 Concrete quotient and observation consequences

The provenance results above determine why the final Health quotient has its particular classes. This section records the inherited quotient consequences and identifies their status as corollaries of Paper III.

Let the complete forest query carrier contain requirement, scenario, and horizon. The state-level Health quotient is

$$q_H : X \rightarrow X/\sim_H, \quad x \sim_H y \iff \forall q, H(q, x) \leftrightarrow H(q, y).$$

Define

$$\text{FullForestHealthCode} = \{\text{allDeclaredHealthy}, \text{contextDependent}, \text{nonRealizing}\}$$

and

$$h_H : X \rightarrow \text{FullForestHealthCode}$$

with fibers $\{B, R, A\}$, $\{P\}$, and $\{F\}$.

The forest calculation proves

$$\boxed{h_H \equiv q_H, \quad \text{FullForestHealthCode} \simeq X/\sim_H.} \quad (48)$$

33

Combining Equation (48) with the inherited Paper-III theorem (1) gives the forest-specific corollary:

Corollary 10.1 (Forest observation sufficiency). *For every observation $M : X \rightarrow O$,*

$$\boxed{\text{HealthObservationSufficient}(H, M) \iff M \succeq h_H.} \quad (49)$$

34

The conceptual content specific to this paper lies upstream of this corollary: the exact provenance analysis explains why the quotient has these fibers and identifies the stage at which each lost distinction disappears.

³³Formal counterpart: `lean/Tier33/Health/Forest02/FullHealthQuotient.lean::fullHealthCode_equivalen
t_healthFamilyQuotient;fullForestHealthCodeEquiv;fullForestHealthCodeEquiv_commutes.`

³⁴Formal counterpart: `lean/Tier33/Health/Forest02/MinimalObservation.lean::forest_observation_suffi
cient_iff_refines_fullHealthCode;forest_observation_sufficient_iff_unique_code_factor.`

11 Additional formal integration checks

The same frozen forest object includes further machine-checked comparisons used as integration checks for Papers II–III. Persistent and intermittent support have equal short-horizon capacity and different long-horizon capacity, so no uniform short-to-long reconstruction map exists on the declared support family.³⁵

A recovery relation can be raw-capacity-indeterminate while retaining a determinate health-visible target; a second branch is indeterminate even after the Health quotient.³⁶

Recovered and adapted have equal original mild/now capacity, but minimum declared stable-response costs 1 and 3. Budget 2 therefore separates them in a cost-enriched representation.³⁷

The benchmark also includes a lawful, continuing canopy-turnover history containing individual-tree mortality and retaining the baseline forest state. This is a finite scale-consistency witness for organization-level continuation amid component turnover.³⁸

Context/stage licensing, control, recovery routes, probability-sensitive adequacy, and failure-aware robustness are retained in the existing formal artifacts. Their supporting roles and theorem map appear in Appendices B and C. None enlarges the main equality-kernel or conservative-refinement claim.

12 Discussion

12.1 The benchmark’s principal lesson

The finite forest is useful because several kinds of information loss occur in one transparent chain and are completely classifiable.

The semantic filters act first. Candidate regeneration at mild/now is removed by lawfulness. Severe/long collapse survives the lawfulness filter and is removed by constitutive continuation. These stages concern which response histories enter prospective capacity at all.

Representation acts next. The detailed observer retains baseline, recovery, and adaptation provenance separately. The original observer maps all three stable modes to one atom. This causes the unique distinct-state collision R/A in the full coarse-capacity profile.

The declared Health language acts after representation. It maps baseline, recovered, and adapted to one full Health class, even though baseline remains distinguishable from the latter two in coarse capacity. This produces the complete query-relative collision family B/R and B/A .

³⁵ *Formal counterpart:* `lean/Tier33/Health/Forest/Support.lean:: persistent_intermit_same_short_forest_capacity; persistent_intermit_split_long_forest_capacity; no_uniform_short_to_long_raw_capacity_map; no_uniform_short_to_long_health_visible_map.`

³⁶ *Formal counterpart:* `lean/Tier33/Health/Forest/Transport.lean:: recoveryBranch_raw_indeterminate; recoveryBranch_healthCode_determinate; recoveryBranch_healthClass_determinate; negativeBranch_healthClass_indeterminate.`

³⁷ *Formal counterpart:* `lean/Tier33/Health/Forest/Cost.lean:: equal_forest_capacity_different_minimum_cost; forest_budget_distinguishes_equal_capacity; forest_budget_query_boundary.`

³⁸ *Formal counterpart:* `lean/Tier33/Health/Forest/ScaleBoundary.lean:: individual_tree_mortality_compatible_with_forest_continuation.`

A shared final Health answer can therefore arise through multiple information histories. The exact localization theorems identify the first stage producing the equality. When a distinction remains in coarse capacity, a new query can use it without reconstructing a richer response representation. When representation has already erased a distinction, every downstream predicate of that representation shares the erasure, and richer information is required.

Information availability and scientifically licensed relevance are distinct obligations. The equality-kernel census proves which finite states can be distinguished by each declared observer or query family. It does not establish whether a newly distinguishable ecological feature ought to enter an empirical Health requirement. In incomplete-data measurement, even an available mathematical distinction may lack reliable observational recovery.

12.2 Conservative evolution of a measurement representation

The generic conservative-refinement lemma records the compatibility condition for evolving representations: exact projection recovers the established representation, and adequacy pullback recovers the established adequacy semantics. The detailed forest observer satisfies these conditions strictly and recovers a distinction erased by the established representation.

The benchmark separates two design obligations: compatibility with an existing declared measurement interpretation and availability of additional information for a new question. The strict forest projection satisfies the first exactly, and its detailed atoms satisfy the second for adaptation mode. Scientific licensing of that new query, or empirical calibration of a corresponding observation, is an additional task.

12.3 Relation to ecological interpretation

The finite labels are formal. Their role is to create an auditable benchmark for information flow through a health specification. Ecological resilience, succession, forest condition, and disturbance-response theory provide neighboring scientific vocabularies [4, 5, 6, 7]. Empirical use of the prospective-health framework would require domain-specific justification of state variables, lawfulness, constitutive realization, continuation criteria, observation maps, adequacy requirements, model uncertainty, and validation.

The representation order used here is an equality-kernel refinement order for deterministic encodings. Blackwell’s ordering of statistical experiments and abstract-interpretation Galois structures provide neighboring formalisms with different comparison structures [8, 9]. Their relationship to the present construction is comparative and contextual: the current paper uses equality-kernel refinement as its operative order.

12.4 What transfers to stochastic measurement, and what does not

An exact projection of set-valued capacity establishes equality of attained response supports under the stated observer compatibility. A stochastic measurement system also assigns a law to its candidate histories. For fine and coarse state and path spaces, write $\pi_X : X_f \rightarrow X_c$ and $\pi_\Gamma : \Gamma_f \rightarrow \Gamma_c$. A

possible *additional* compatibility condition is the pushforward identity

$$\boxed{(\pi_\Gamma)_\# \mu_f(x; d, t) = \mu_c(\pi_X x; d, t).} \quad (50)$$

This is a candidate condition for stochastic implementations, not a theorem established by the frozen Lean artifact. For corresponding viable sets V_f, V_c , if additionally

$$\mathbf{1}_{V_f}(\gamma) = \mathbf{1}_{V_c}(\pi_\Gamma \gamma) \quad \mu_f\text{-almost surely}, \quad (51)$$

then the usual pushforward identity gives the conditional consequence

$$\mu_f(V_f) = \mu_c(V_c), \quad \mu_f(\Gamma_f \setminus V_f) = \mu_c(\Gamma_c \setminus V_c). \quad (52)$$

Preservation of total viable mass alone leaves reserve-qualified viability undetermined. For a reserve threshold ρ , let $R_f^\rho \subseteq \Gamma_f$ and $R_c^\rho \subseteq \Gamma_c$ be the measurable events that a history retains the required terminal reserve. In addition to the path-law pushforward and viability compatibility above, suppose that reserve qualification is preserved under the path projection:

$$\boxed{\mathbf{1}_{R_f^\rho}(\gamma) = \mathbf{1}_{R_c^\rho}(\pi_\Gamma \gamma) \quad \mu_f(x; d, t)\text{-almost surely}.} \quad (53)$$

Then the intersection of viable and reserve-qualified histories has the same original-law mass in both systems:

$$\boxed{\begin{aligned} m_{R,f}(\rho) &:= \mu_f(x; d, t)(V_f \cap R_f^\rho) \\ &= \mu_c(\pi_X x; d, t)(V_c \cap R_c^\rho) =: m_{R,c}(\rho). \end{aligned}} \quad (54)$$

For the common inclusive threshold θ , the continuation query $Q_1 = [m_V \geq \theta]$ and reserve-qualified query $Q_2 = [m_R(\rho) \geq \theta]$ therefore agree across the projection. If present realization is also compatible, $\text{Realizes}_f(x) \iff \text{Realizes}_c(\pi_X x)$, their corresponding Health judgments agree. These are elementary conditional consequences of the stated measure and event compatibilities. Application to observed response distributions additionally requires compatible observation maps and declared lawfulness and continuation semantics. No stochastic-compatibility hypothesis in this paragraph is asserted for Paper V's independently constructed generators or proved by the frozen Lean artifact.

The companion study provides two deliberately different enrichment cases. First, its $M_3 \rightarrow M_2$ forgetting map removes a deterministic density coordinate computed from information already retained by M_2 ; every tested Q1/Q2 answer agrees under that projection on the same fixed response bank. This is an exact, query-redundant compatibility observation for the declared computations, without a claim of a nontrivial gain or preservation for arbitrary adequacy. Second, juvenile-size enrichment preserves initial count and basal area to numerical tolerance, but the coarse and enriched stochastic transition implementations have no proved projection-commutation or common coupling. Their downstream capacity differences are conditional state/model comparisons; the exact preservation theorem here makes no prediction for them without its compatibility hypotheses.

13 Formal and empirical measurement: complementary results

The finite forest benchmark and the empirical-synthetic forest study address different scientific questions. Table 2 makes their respective ownership and limitations explicit. The companion’s original-law weighting produces a viable subprobability measure, with unlawful and lawful-noncontinuing mass in a separate complementary accounting ledger; it does not insert failure-history atoms into the viable capacity.

Interface	Paper IV: exact formal benchmark	Paper V: empirical-synthetic investigation
Histories and admissibility	Candidate/lawful/viable carriers, filtration, and history-encoding equivalence	Generated paths; model-admissibility checks; stipulated continuation
Observation and representation	Equality kernels; exact location of lost or query-invisible distinctions	Partial reconstruction calibration; juvenile-size information loss
Capacity	Set of attained viable response atoms	Unrenormalized viable subprobability measure; unlawful and lawful-noncontinuing mass tracked in a separate complementary ledger
Query-redundant enrichment	Exact preservation when the theorem’s capacity-projection hypotheses hold	$M_3 \rightarrow M_2$ drops deterministic density information and preserves the tested Q1/Q2 answers on the fixed response bank
Size-sensitive enrichment	Exact projection/pullback guarantees require shared dynamics and constitution	Juvenile-size enrichment preserves initial count and basal area, while future-kernel projection compatibility remains unproved
Adequacy	Exact predicates on declared capacities	Threshold decisions exposed to estimation error, misspecification, and numerical uncertainty
Scientific authority	Consequences of fully declared formal semantics	Empirical identification, ecological licensing, and prospective-validation obligations

Table 2: Independent contributions of the exact benchmark and its empirical-synthetic companion. Neither verification procedure establishes the other’s premises.

The term *lawfulness* names different declared obligations in the two studies. In the finite benchmark, **Lawful** filters candidate histories according to contextual and horizon-relative admissibility, including the stage at which regeneration is allowed. In Paper V, **Lawful_G** verifies integrity of paths produced by the numerical generator, including nonnegative counts, admissible diameters, and population accounting. Satisfying that integrity predicate leaves ecological licensing and calibration of the transition law as independent scientific obligations.

Paper V exposes unidentified candidate-entry intensity, incomplete individual-size recovery, omitted species-specific dynamics, query-relevant RMS compression, and sensitivity of threshold judgments to decision margin and context information. Its controlled synthetic tests distinguish favorable broad-range discrimination from weaker discrimination near the adequacy boundary. Two completed diagnostics make the distinction between query-level and model-level adequacy concrete: a selected one-year reconstruction comparison localizes all nonviable candidate paths to terminal juvenile-support failure, while independent adult-mortality observations place hemlock deaths far outside the stipulated shared-hazard predictive distribution. The former locates a failure in a declared continuation predicate; the latter tests the empirical adequacy of a specified stochastic law.

Both conclusions retain their respective experimental and observational supports. The study does not identify an operational Health status for Harvard Forest or establish the demographic generator as the correct ecological law.

These observations provide an empirical test of the *obligations* identified by the finite benchmark, rather than proof that the two implementations are equivalent. Paper IV proves exact semantic invariance and preservation under verified hypotheses; Paper V documents which observation, representation, transition-law, and numerical obligations can be met in a concrete data-anchored experiment and which remain limited.

14 Scope boundaries

Formal completeness. All forest-specific theorems are exact within the declared five-state carrier, history constructors, candidate-lawfulness predicate, contexts, horizons, observers, and query family. The equality-kernel ladder is a complete classification *of these finite encodings*. The five labels are not asserted to form a general taxonomy of ecological states or empirical Health.

Representation completeness. The original and detailed capacities are set-valued attained-response representations. Their equality kernels determine distinguishability in the declared formal model. Neither that exhaustiveness nor the observation-sufficiency corollary supplies statistical identifiability of latent ecological states from incomplete observations. Such recovery requires an independently justified observation kernel and calibration evidence.

Dynamic compatibility. The conservative-refinement theorem assumes the same dynamics and constitution and an exact capacity projection. It does not automatically transfer across independently changed predictive laws, reconstructed state spaces, or transition mechanisms. The stochastic pushforward and semantic-commutation requirements discussed in Section 12.4 are explanatory compatibility obligations, not already verified theorems of the frozen formal artifact.

Scientific licensing. Lean checks formal consequences of declared datatypes, definitions, and proof terms. The appropriateness of realization predicates, response atoms, costs, scenarios, and adequacy requirements for an empirical forest remains a scientific question. The companion study addresses portions of the observation, reconstruction, and numerical tasks while explicitly retaining unidentified processes and unresolved prospective validation.

The distinction between formal specification and scientific licensing is therefore material. The finite theorem provides exact reference conditions; ecological and statistical applicability requires separate evidence.

15 Conclusion

This paper establishes a machine-checked reference model for the information architecture of prospective Health. Two equivalent history encodings relocate contextual admissibility between dependent constructors and a nontrivial lawfulness filter while preserving viable continuation, observed capacity, and Health. A complete finite response census then separates candidate histories, lawful histories, and constitutively viable continuations.

Detailed capacity, original capacity, and the complete declared Health language induce the exact equality-kernel ladder

$$\boxed{B \mid P \mid R \mid A \mid F \longrightarrow B \mid P \mid \{R, A\} \mid F \longrightarrow \{B, R, A\} \mid P \mid F.}$$

The classification identifies every distinction erased by capacity representation and every additional distinction retained by capacity but unused by the declared Health language. These diagnoses have different measurement consequences: new queries can use retained information, whereas erased distinctions require a richer representation or new evidence.

The generic conservative-refinement lemma identifies an exact capacity projection and adequacy pullback as sufficient conditions for preserving established Health judgments. Its strict forest instance proves that additional response detail can expose an adaptation-sensitive distinction while preserving arbitrary established adequacy. The result is a modest, reusable compatibility guarantee with an independently checked nontrivial example.

These statements concern fully declared finite semantics and set-valued attained-response capacity. Probability-weighted capacity, stochastic model compatibility, and scientific calibration require additional assumptions and evidence. The companion empirical-synthetic measurement study investigates these obligations using incomplete census support, provisional demographic dynamics, and threshold-dependent numerical decisions. Together, the two papers separate guarantees supplied by a formal measurement architecture from reliability established through empirical measurement and validation.

A Formal source and verification

The formal development for this manuscript is distributed across three frozen forest tranches:

Surface	Commit
FOREST01	2ba13ec536255b4be76ddd25d7e5b1ce59f0d23a
FOREST02	82d10131bb30e338a6267b3ec8f978a8dbc07800
FOREST03	f4b9bff54ffcc55a1271a67a11c84c3b657f1a1f

Preferred import roots are

`Tier33.Health.Forest,` `Tier33.Health.Forest02,` `Tier33.Health.Forest03.`

The generic conservative-refinement result is owned by

`Tier33.Health.Representation.ConservativeRefinement.`

Verification commands are

```
python3 scripts/verify_t33_health_forest01.py -compile
python3 scripts/verify_t33_health_forest02.py -compile
python3 scripts/verify_t33_health_forest03.py -compile
```

In the previously recorded formal verification, the FOREST03 gate passed with 899 frozen predecessor and dependency-pin owner files byte-identical, 143 exact authored/constructor/projection names checked, 30 completion fields kernel-checked as propositions, eight independent universe levels in the generic conservative-refinement schema, and ten hostile false claims rejected for their intended mathematical reasons. The 6,044-module local import cut required 48 private outputs; 6,003 shared artifact digests remained unchanged. The axiom audit reported the inherited standard dependencies `propext`, `Quot.sound`, and `Classical.choice` where required.

The formal development uses the repository’s pinned Lean and Mathlib environment; the exact dependency locks are carried by `lean-toolchain` and `lake-manifest.json`. The working formal repository is private. The results above refer to the recorded prior FOREST01–03 verification; this editorial revision changes no Lean source and makes no claim of a new Lean compilation or hostile-control run. The Paper-IV public artifact and its permanent locator remain pending separate source-disclosure and archive preparation. The companion Paper-V computational release is distinct and remains subject to its own disclosure, licensing, and public-deposition process.

B Selected theorem map

ID	Role	Principal formal counterpart
P4-01	Original forest capacity census	Forest/Core.lean::capacity_eq_expected
P4-02	Original Health census and context effects	Forest/HealthLanguage.lean::forest_health_census;requirement_changes_forest_health;scenario_changes_forest_health;horizon_changes_forest_health
P4-03	Nontrivial candidate lawfulness	Forest03/CandidateSemantics.lean::candidate_regenerate_now_not_lawful;candidate_regenerate_long_lawful;candidate_collapse_mild_not_lawful;candidate_collapse_severe_lawful
P4-04	Lawful-history encoding equivalence	Forest03/LawfulEncodingEquivalence.lean::lawfulCandidateEquivHistory;candidate_history_roundtrip;original_history_roundtrip
P4-05	Viable-continuation encoding equivalence	Forest03/LawfulEncodingEquivalence.lean::candidateViableEquivOriginal;candidate_viable_roundtrip;original_viable_roundtrip
P4-06	Capacity and Health encoding invariance	Forest03/LawfulEncodingEquivalence.lean::candidate_capacity_eq_original;candidateHealth_iff_originalHealth
P4-07	Complete candidate/lawful/viable response census	Forest03/ResponsePipeline.lean::responseKinds_complete_census;responseKinds_inclusions
P4-08	Strict semantic filters	Forest03/ResponsePipeline.lean::regeneration_removed_by_lawfulness;collapse_removed_by_constitution
P4-09	Viable responses map exactly to detailed and coarse capacity	Forest03/ResponsePipeline.lean::viable_responseKinds_map_detailed_capacity;viable_responseKinds_map_original_capacity
P4-10	Detailed-profile equality kernel	Forest03/InformationLadder.lean::detailedProfile_eq_iff_state_eq
P4-11	Coarse-profile equality kernel	Forest03/InformationLadder.lean::coarseProfile_eq_iff
P4-12	Full Health equality kernel	Forest02/FullHealthCode.lean::fullHealthSignature_eq_iff_code_eq
P4-13	Exact refinement ladder and strictness	Forest03/InformationLadder.lean::information_ladder_refines;coarseProfile_does_not_refine_detailedProfile;healthSignature_does_not_refine_coarseProfile
P4-14	Complete representation-loss classification	Forest03/CollapseLocalization.lean::detailed_to_coarse_collapse_iff;distinct_coarse_collision_iff
P4-15	Complete query-relative loss classification	Forest03/CollapseLocalization.lean::coarse_to_health_collapse_iff;query_invisible_retained_distinction_iff
P4-16	Generic conservative refinement	Representation/ConservativeRefinement.lean::ConservativeCapacityRefinement;ConservativeCapacityRefinement.health_iff;ConservativeCapacityRefinement.capacity_refines
P4-17	Forest conservative-refinement instance	Forest03/ConservativeRefinement.lean::forestConservativeRefinement;forest_arbitrary_adequacy_preserved;forestHealth_preservation_from_generic
P4-18	Strict new-query boundary	Forest02/EnrichmentBoundary.lean::forest_representation_enrichment_boundary;original_capacity_insufficient_for_adaptationMode;detailed_capacity_sufficient_for_adaptationMode
P4-19	Concrete full Health quotient	Forest02/FullHealthQuotient.lean::fullForestHealthCodeEquiv;fullForestHealthCodeEquiv_commutates
P4-20	Forest observation corollary	Forest02/MinimalObservation.lean::forest_observation_sufficient_iff_refines_fullHealthCode;forest_observation_sufficient_iff_unique_code_factor
P4-21	Semantic provenance completion certificate	Forest03/Completion.lean::ForestSemanticProvenanceCertificate;forest_semantic_provenance_complete

C Additional integration results

The frozen forest development also contains machine-checked instances for:

- contextual stage licensing and stage/horizon separation;
- raw-indeterminate and health-visible-determinate relation transport;
- open-system support and short-to-long reconstruction obstructions;
- controlled restoration and endogenous recovery;
- cost-sensitive budget Health and minimum-cost quotient realization;
- recovery-route multiplicity and declared route-class queries;
- probability support, scalar success probability, and threshold adequacy;
- failure-aware robustness using positive and negative challenge profiles;
- component-turnover consistency at forest scale.

These results provide integration coverage of the broader architecture. The central provenance and conservative-refinement results are established by the semantic pipeline, information ladder, collapse-localization, and enrichment modules.

D Series boundary

Paper I owns the prospective-health definition, viable-continuation construction, and adequacy-visible capacity quotient. Paper II owns contextual licensing and exact relation-generated transport. Paper III owns the generic equality-kernel representation order, principal query-sufficiency theorem, single-query boundaries, least sufficient deterministic transport target, attained-range quotient realizations, and the generic state-level observation criterion.

Paper IV owns the finite forest semantic-provenance benchmark, the equivalent permissive lawfulness encoding, the complete candidate/lawful/viable response census, the exact detailed/coarse/Health information ladder, complete localization of both information-loss stages, and the strict conservative-enrichment instance. It also records a generic conservative-capacity-refinement preservation lemma for reuse.

Paper V supplies an independently specified empirical-synthetic measurement study with partial observation-based calibration, model-discrepancy tests, numerical decision diagnostics, and explicit limits on ecological interpretation. Its stochastic history law and executable replay belong to a separate computational artifact. An operational forest Health assessment and prospective ecological validation remain further scientific obligations.

References

- [1] Z. James. Constitutive Continuation Capacity: A machine-checked framework for prospective health. Preprint, 2026. doi:10.5281/zenodo.23131158.
- [2] Z. James. Prospective Health Across Contexts: Contextual Licensing, Developmental Transport, and Open-System Support. Preprint, 2026. doi:10.5281/zenodo.23146281.
- [3] Z. James. Representation Sufficiency in Prospective Health: Query-Visible Quotients, Transport, and Enriched Response Semantics. Preprint, 2026. doi:10.5281/zenodo.23219588.
- [4] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973.
- [5] L. H. Gunderson. Ecological resilience—in theory and application. *Annual Review of Ecology and Systematics*, 31:425–439, 2000.
- [6] S. A. Levin. The problem of pattern and scale in ecology. *Ecology*, 73(6):1943–1967, 1992.
- [7] C. D. Oliver and B. C. Larson. *Forest Stand Dynamics*. Updated edition, John Wiley & Sons, 1996.
- [8] D. Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi:10.1214/aoms/1177729032.
- [9] P. Cousot and R. Cousot. Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints. In *Proceedings of the 4th ACM SIGACT-SIGPLAN Symposium on Principles of Programming Languages*, pages 238–252, 1977. doi:10.1145/512950.512973.
- [10] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science **12699**, Springer (2021) 625–635.
- [11] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, ACM (2020) 367–381.